



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS603 / 20CS702 / 17CS603 - COMPILER DESIGN

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Show the various phases of compiler with neat diagram.	6	L*2	1	1
b)	Derive LMD and RMD for the given string using below grammar and also draw the parse tree. The string is id+id*id The grammar is $E \rightarrow E+E E^*E -E (E) id$ Check whether the grammar is ambiguous or not.	8	L3	2	1,2
c)	With example explain the concept of input buffering Lexical analysis.	6	L3	1	1,2
2. a)	Construct the transition diagram for recognizing the tokens given below i. Relational operator ii. Unsigned numbers	6	L3	2	1,2
b)	Explain the applications of compiler technology.	4	L2	1	1
c)	Write the algorithm for the generation of FIRST() and FOLLOW() sets.	6	L2	2	1
d)	Calculate FIRST() and FOLLOW() for the given grammar. $S \rightarrow A$ $A \rightarrow aB / Ad$ $B \rightarrow b$ $C \rightarrow g$	4	L3	2	1
3. a)	Construct LL(1) parsing table for the given grammar and parse the given string: id + id *id. The grammar is $E \rightarrow E+T T$ $T \rightarrow T * F F$ $F \rightarrow (E) id$	10	L3	2	1,2
b)	Eliminate left recursion in the following grammar i) $S \rightarrow A$ $A \rightarrow Ad Ac aB aC$ $B \rightarrow bBc f$ ii) $A \rightarrow ABd / Aa a$	4	L3	2	1,2
c)	Write a LEX program to recognize valid arithmetic expression. Identifiers in the expression could be only integers and operators could be + and *. Count the identifiers and operators present and print them separately.	6	L3	1	1
Unit – II					
4. a)	Construct SLR(1) parsing table for the given grammar and parse the input string id*id+id $E \rightarrow E + T T$ $T \rightarrow T * F F$ $F \rightarrow (E) id$	10	L4	3	1,2

P.T.O.

- b) Write the syntax Directed Definition for simple calculator and draw annotated parse tree for the given expression: $3*5+4n$. 10 L3 4 1,2
5. a) Construct CLR(1) parsing table for the given context free grammar
 $S \rightarrow AA$
 $A \rightarrow aA|b$ 10 L4 3 1,2
- b) Explain Quadruple, Triple and indirect triple. Translate the following expression, into Quadruple, Triple and indirect triples $b* - c + b* - c$. 10 L3 4 1,2
6. a) Construct LALR(1) parsing table for a given grammar
 $S \rightarrow Aa|bAc|dc|bda$
 $A \rightarrow d$
 Parse input string "bdc" using LALR parsing table. 10 L4 3 1,2
- b) Construct syntax tree for the expression $a - 4 + c$ using grammar.
 $E \rightarrow E + T$
 $E \rightarrow E - T$
 $E \rightarrow T$
 $T \rightarrow (E)$
 $T \rightarrow id$
 $T \rightarrow num$ 10 L3 4 1,2
- Unit – III**
7. a) Describe the various issues in the design of a code generator. 10 L2 5 1
- b) Explain the code generation for the arithmetic operation and write the code for the following statement.
 $t := a - b$
 $u := a - c$
 $v := t + u$
 $d := v + u$ 10 L3 5 1
8. a) What are basic blocks? How do you partition a three address code into basic blocks? Explain. 10 L2 5 1
- b) What is the need for optimization? With an example, explain different code optimization techniques. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
